

Aswin VATTAPPARAMBATHU JAYAPRAKASH

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XR Developer and Researcher specializing in Unity-based immersive experiences, multisensory feedback systems (thermal, haptic, spatial audio), and AI-powered computer vision applications. Proven track record developing cross-platform VR/AR/MR applications using OpenXR, deployed on Meta Quest, HTC Vive, and Pico headsets. Open-source contributor with merged pull requests to research frameworks.

SKILLS

Programming languages & Frameworks: Python, C++, C#, Scilab (Matlab alternative), Unity (C#), WebXR, Three.js, PyTorch, Keras, ROS 2, SLAM

XR/Graphics: OpenXR, XR Interaction Toolkit, Meta Quest SDK, GLSL Shader Programming, WebGL, Real-time 3D Visualization

Tools: Git/GitHub, CUDA, LaTeX, DOTween, DaVinci Resolve

Languages: English (Bilingual), Hindi (Bilingual), Malayalam (Native), French (Beginner), Japanese (Beginner)

EDUCATION

IMLEX - MSc Engineering, Lighting and XR | MSc in Optics, Image, Computer Vision, Machine Learning and Multimedia | MSc of Computer Science • 09/2024 -

University of Eastern Finland | Universite Jean-Monnet | Toyohashi University of Technology • Finland | France | Japan 09/2026

- Multidisciplinary program combining image processing, lighting design, and XR development across 3 institutions
- Master's Thesis (Upcoming): "Industrial Training Using VR/AR-Based Prototyping and AI" at National Institute of Advanced Industrial Science and Technology (AIST), Japan (Mar–Aug 2026)

BSc (honors) in Physics • Hindu College, University of Delhi • Delhi, India 11/2020 - 07/2023

WORK EXPERIENCE

XR Research and Development Intern

LIRIS (Laboratoire d'InfoRmatique en Image et Systèmes d'information) • Saint-Étienne, France 06/2025 - 09/2025

- Enhanced open-source 6DoF VR Thermal Feedback System (post-IEEE VR 2025 publication) by implementing obstacle-aware thermal rendering using Unity raycasting and collision detection; merged contributions to main repository, improving system realism for future XR haptics research
- Integrated Skinitic Haptic Vest with custom vibrotactile feedback patterns synchronized to gameplay events (rain simulation with distributed torso pulses, vehicle close-passes with lateral vibrations, collision feedback with impact pulses)
- Developed multisensory VR demo combining thermal lamps, haptic vest, and spatialized audio (via ASIO Audio Manager plugin) in a road-crossing scenario; user testing confirmed enhanced immersion compared to baseline VR-only experience
- Improved developer experience by adding intuitive material system (color-coded thermal sources), teleport navigation, and hit-feedback visualization to demo scenes

Student Research Assistant

Hindu College, University of Delhi • Delhi, India 06/2023 - 01/2024

- Conducted comprehensive literature review of laser-based automotive headlamp technologies, analyzing luminous efficiency, ADB capabilities, and thermal management; identified key engineering challenges (heat dissipation, laser safety, cost optimization) to inform future research directions

PROJECTS

Plank Experiment: VR Perception Study with Physical Props 11/2025 - 11/2025

Toyohashi University of Technology, Japan

[Click to view project]

- Developed Unity-based experimental framework for XR perception research investigating how physical realism influences height perception and fear responses in VR (team of 4, responsible for all VR interactions and data collection systems)
- Engineered randomized trial system with elevator mechanics and height-adjustable plank scenarios (5 height levels × 4 trials × 2 conditions = 40 trials per participant)
- Implemented automated data logging for subjective height judgments, movement timing, and behavioral metrics; results showed physical plank lowered Point of Subjective Equality (PSE) and increased cautious movement time

VR Sound Visualizer: Accessible WebXR Experience for Deaf/Hard-of-Hearing Users

05/2025 - 06/2025

Jean Monnet University, France

[Click to view project]

- Architected WebXR application using Three.js and JavaScript that converts non-verbal audio (birdsong, ambient sounds) into 3D directional indicators and real-time spectrograms for accessibility
- Ideated and integrated multisensory feedback system combining visual audio spectrum (pitch/volume), directional proximity indicators, and haptic controller vibrations based on sound source distance
- Optimized WebGL rendering pipeline for smooth performance on VR headsets; implemented JSON-based configuration system for environment customization

Face-to-Ball: Deep Learning-Based Lighting Transfer Using Synthetic Data

03/2025 - 05/2025

Jean Monnet University, France

[Click to view project]

- Synthesized photorealistic dataset of 1,600 paired samples (16 face models × 100 lighting/viewpoint conditions) using 3D reconstruction tools (KeenTools FaceBuilder, TRELIS AI) and consistent rendering pipeline
- Designed and trained U-Net architecture with ResNet34 encoder for facial relighting tasks; implemented hybrid loss function (MSE + SSIM) achieving 15% SSIM improvement over 500 training epochs
- Validated end-to-end pipeline demonstrating successful lighting transfer (shadows, highlights, light direction) from facial images to reference sphere

Real-Time 3D Color Cloud Visualization (WebGL & Three.js)

02/2025 - 02/2025

Jean Monnet University, France

[Click to view project]

- Designed and implemented custom GLSL shaders for rendering RGB, CIExyY, and CIELAB color spaces in 3D point clouds
- Developed VR/MR interactions for Meta Quest allowing users to manipulate video feeds and color clouds with 6DoF controllers
- Built real-time processing pipeline handling video input and color space transformations at 60fps

***Additional projects and professional experience available at:**

aswin-avj.github.io/portfolio

VOLUNTEERING & LEADERSHIP

Panchtatva, Environment Society

01/2022 - 01/2023

Media Head • Delhi

- Managed video production and graphic design for sustainability campaigns reaching 10,000+ audience members

Eureka, Physics Society

01/2022 - 01/2023

Head Member • Delhi

- Led video creation for AI and astronomy workshops; co-organized scientific literacy events